KLE Technological University | Creating Value, Leveraging Knowledge

Department of Computer Science and Engineering (Artificial Intelligence)

\FMTH0306-1.0

Course Plan

Semester: V **Year: 2026-27**

Course Title: Cloud Computing		Course Code: 26ECAC301
L-T-P:3-0-1	Credits: 4	Contact Hrs: 5 hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 40	Tutorial/Practical: 30hrs	Exam Duration: 3 hrs
Lesson Plan Authors: Sadaf .S, Dheeraj Kumar		Date: 01/08/2026
Checked By: Dr. Narayan D G		Date: 08/08/2026

Prerequisites: Fundamental concepts of Computer Networks

Course Outcomes (COs):

At the end of the course, the student should be able to:

1. Apply cloud services and delivery models to deploy cloud- based applications.
2. Analyze virtualization and containerization technologies and their performance.
3. Demonstrate various levels of automation and container orchestration using Kubernetes
4. Apply Kubernetes to deploy and manage microservices and serverless applications across cloud and edge environments.
5. Analyze and optimize resource management in the edge–cloud continuum through application placement, computation offloading and load balancing.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Cloud Computing													Semester: V
Course Code: 26ECAC301													Year: 2026-27
Course Outcomes (COs) / Program Outcomes (POs)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
1. Apply cloud services and delivery models to deploy cloud- based applications.	H												
2. Analyze virtualization and containerization technologies and their performance.	M	H			H								
3. Demonstrate various levels of automation and container orchestration using Kubernetes			M		H							M	
4. Apply Kubernetes to deploy and manage microservices and serverless applications across cloud and edge environments.	M				H							H	
5. Analyze and optimize resource management in the edge–cloud continuum through application placement, computation offloading, and load balancing.		M			H								

Degree of compliance **L**: Low **M**: Medium **H**: High

Competency addressed in the Course, corresponding Performance Indicators (PI) and Course Specific Performance Indicators (CSPI)

Competency	Performance Indicators (PI)	Course Specific Performance Indicators (CSPI)
1.4: Demonstrate the competency in Computer Science and Engineering (WK4)	1.4.3: Apply network-engineering, security, and Cloud Computing principles to build secure infrastructures, implement secure code, assess vulnerabilities, and design risk-based frameworks.	1.4.3.1: Apply cloud services and deployment models for cloud-based applications.
		1.4.3.2: Apply various cloud service models (IaaS, PaaS, SaaS) to determine the most suitable model for a given application.
2.1: Demonstrate an ability to identify and define complex engineering problems by understanding real-world contexts and constraints.	2.1.2: Identify critical systems/subsystems, variables, and constraints that characterize the problem environment by analyzing practical contexts and real-world limitations.	2.1.2.1: Identify various virtualization techniques and types of hypervisors used in cloud environments.
		2.1.2.2: Understand the operations of virtualization and containerization, including VM creation, container execution environments, and runtime abstraction.
3.2: Demonstrate an ability to generate a diverse set of alternative design solutions.	3.2.2: Identify suitable criteria for the evaluation of alternate design solutions.	3.2.2.1: Build container orchestration using Kubernetes for a web application.
		3.2.2.2: Build Kubernetes clusters and implement scheduling, resource allocation, and workload placement across nodes.
	3.2.3: Refine a conceptual design into a detailed design within the existing constraints.	3.2.3.1: Evaluate cloud, edge, and distributed applications using performance metrics such as latency, bandwidth usage, resource utilization, and response time.
		3.2.3.2: Evaluate automation approaches using infrastructure-as-code and automation tools.
5.2: Demonstrate an ability to apply domain-specific tools, modelling techniques, and resources.	5.2.1: Create/modify parts and models using modern engineering and IT tools for the solution of complex engineering problems.	5.2.1.1: Apply configuration management to reduce manual interventions.
		5.2.1.2: Apply automation practices for streamlining cloud infrastructure and application deployment.
12.2: Ability to deploy reliable software systems through systematic testing and validation.	12.2.2: Deploy a reliable software system by identifying appropriate resources.	12.2.2.1: Deploy appropriate cloud service models for a web-based application.
		12.2.2.2: Deploy containerized applications using suitable containerization and orchestration mechanisms.

Eg: 1.2.3.4: Represents program outcome ‘1’, competency ‘2’, performance indicator ‘3’ and Course Specific Performance Indicator ‘4’’.

Course Content

Course Code: 26ECAC301	Course Title: Cloud Computing	
L-T-P: 3-0-1	Credits: 4	Contact Hrs:4hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 40	Practical Hrs:30 Hrs	Exam Duration: 3 hrs

Content	Hrs.
Unit - I	
Chapter 1: Introduction Motivation for cloud computing, elastic computing and scalability, centralized and distributed computing models, cloud service models, cloud deployment models, edge computing motivation, role of edge in reducing latency and bandwidth usage, data center architecture, virtualization overview, cost and performance trade-offs in cloud and edge systems.	8 hrs
Chapter 2: Virtualization and containerization Virtual machines and virtualization approaches, types of hypervisors and virtualization models, containerization concepts, operating system isolation mechanisms, container execution environment and runtime abstraction, container networking concepts, persistent storage and data management for containers, comparison of virtual machines and containers and associated trade-offs.	7 hrs
Unit – II	
Chapter 3: Automation and Orchestration Orchestration principles, Kubernetes architecture, cluster model including nodes, pods, and control plane, container lifecycle management, scheduling and resource allocation, workload placement across nodes, Kubernetes-based edge architectures, infrastructure as code concepts and automation tools.	8 hrs
Chapter 4: Microservices and Distributed Applications Microservices Architecture, Advantages and Limitations of Microservices, Service Granularity, Service Discovery, Load Balancing, Stateless Servers and Containers, Development and Deployment of Microservices, Cloud and Edge Deployment of Microservices, Challenges in Distributed System Design, Serverless Computing Approach, and Serverless Infrastructure Architecture.	7 hrs
Unit-III	
Chapter 5: Edge–Cloud Continuum Edge computing fundamentals, edge–cloud continuum, device–edge–cloud hierarchy, latency-sensitive and real-time applications, application placement strategies, computation offloading basics, data locality and communication overhead, latency-aware system design, use cases in AI, IoT, and real-time systems.	5 hrs
Chapter 6: Resource Management Resource allocation strategies, task scheduling, load balancing, performance evaluation metrics and system optimization strategies, optimization strategies for distributed systems, large-scale system case studies.	5 hrs

Text Books:
1. Douglas Comer, "The Cloud Computing: The Future of Computing", 1st ed, Chapman and Hall/CRC 1 July 2021.
2. Dan C. Marinescu, Cloud Computing Theory and Practice, 3rd Edition, Elsevier - February 15, 2022.
Reference Books:
1. Rajkumar Buyya, Christian Vecchiola, S .ThamaraiSelvi, Mastering Cloud Computing, McGraw Hill, 2013.
2. Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, Cloud Computing, A Practical Approach, McGraw Hil, 2010.

Evaluation Scheme

ISA Scheme

Assessment	**Conducted for marks**	**Weightage in Marks**
ISA-1 (Theory)	40	33
ISA-2 (Theory)	40	
Lab Activity	10	17
Course Project	10	
Total		**50**

ESA Scheme

Assessment	**Conducted for marks**	**Weightage in Marks**
Theory	100	33
Lab	50	17
Total		**50**

Course Unitization for ISA and ESA

Topics / Chapters	Teaching Credits	No. of Questions in ISA-1	No. of Questions in ISA-2	No. of Questions in Activity	No. of Questions in ESA
Unit I					
Chapter 1: Introduction.	08 hrs	1	--	0	1
Chapter 2: Virtualization and Containerization.	07 hrs	1	--	1	1
Unit II					
Chapter 3: Automation and Orchestration.	08 hrs	--	1	1	1
Chapter 4: Microservices & Distributed Applications	07 hrs	--	1	1	1
Unit III					
Chapter 5: Edge-Cloud Continuum	05 hrs	--	--	1	1
Chapter 6: Resource Management	05 hrs	--	--	1	1

Note
1. Each Question carries 20 marks and might consist of sub-questions.
2. Mixing of sub-questions from different chapters within a unit (Unit I , Unit II and Unit III) is allowed in ISA I, ISAII, and ESA.
3. Answer 4 full questions of 20 marks each (two full questions from Unit I, and two full questions from Unit II) out of 8 questions in ESA.

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Course Assessment Plan

Course Title: Cloud Computing **Code:** 26ECAC301

Course outcomes (COs)	Weightage in assessment	Assessment Methods			
		ISA I	ISA II	Lab Assessment	ESA (Theory)
1. Apply cloud services and delivery models to deploy cloud-based applications.	20%	✓		✓	✓
2. Apply virtualization and containerization techniques for resource management in Cloud environment.	20%	✓		✓	✓
3. Demonstrate various levels of automation and container orchestration using Kubernetes	20%	✓		✓	✓
4. Develop scalable cloud applications using microservices and DevOps practices for automated deployment, configuration management, infrastructure provisioning, and IT operations optimization.	20%		✓	✓	✓
5. Analyze and optimize resource management in the edge–cloud continuum through application placement, computation offloading, task scheduling, and load balancing.	20%		✓	✓	✓
Weightage	**100%**	**33%**	**33%**		**34%**

Course Activity and Rubrics:

Credit: 1	**Cloud Computing Lab**
Objective:	The student should be able to apply Cloud Computing concepts and tools for virtualization, containerization, orchestration, microservices, edge-cloud computing and resource management.
Team size:	2-3 students (group activity for selected experiments; individual assessment for other experiments).
Type:	Each batch will work on distinct Cloud Computing application and deployment scenarios.

Sl. No.	Experiments	CO	Blooms level	Timeline w.r.t COE	PI code	Hrs	Marks
1.	Performance analysis of Type-1 and Type-2 hypervisors.	CO1	L3	1st Week	1.4.3	2	Nil
2.	Performance evaluation of virtual machines and containers.	CO2	L3	2nd Week	2.1.2	2	Nil
3.	Build, deploy and analyze a containerized application under varying workloads.	CO2	L4	3rd & 4th Week	2.1.2	3	10
4.	Analyze resource allocation, scheduling and load balancing in Kubernetes.	CO3	L4	5th & 6th Week	3.2.2	3	10
5.	Build and deploy a microservices-based web application with service discovery and load balancing.	CO3	L4	7th Week	3.2.2	2	Nil
6.	Deploy and evaluate Serverless framework on kubernetes	CO4	L4	8th–10th Week	3.2.3	4	10
7.	Deploy and evaluate an AI application across cloud and edge environments.	CO5	L4	11th & 12th Week	3.2.3	3	10
8.	Analyze AI workload placement and dynamic computation offloading across device, edge and cloud.	CO5	L4	13th–15th Week	5.2.1	5	10
Total							50

Rubrics for Lab Evaluation

Parameter (evaluated for 10 M)	Excellent (8-10)	Good (4-7)	Fair (1-3)	PI	CO
Containerized Application Deployment	· Successfully builds and deploys a containerized application. · Effectively configures container images, networking, and persistent storage. · Demonstrates proper performance analysis under varying workloads.	· Builds and deploys the containerized application with minor configuration or performance issues. · Demonstrates basic workload analysis.	· Unable to properly build or deploy the application. · Limited understanding of container configuration and performance analysis.	1.4.3	2
Cluster Deployment and Resource Management	· Successfully deploys an application in a cluster-based environment. · Effectively demonstrates resource allocation, scheduling, and workload placement. · Clearly analyzes resource utilization and performance.	· Deploys the application with minor issues. · Demonstrates basic resource allocation and scheduling concepts.	· Unable to deploy or manage the application in the cluster. · Limited understanding of resource allocation and scheduling.	3.2.2 / 12.2.2	3
Microservices-Based Application	· Successfully designs and deploys a well-structured microservices-based application. · Effectively demonstrates service discovery and load balancing. · Clearly explains communication among distributed services.	· Implements the microservices application with minor limitations. · Demonstrates basic service discovery or load balancing.	· Unable to properly implement or deploy microservices. · Limited understanding of service discovery and load balancing.	3.2.3 / 5.2.1	4

Cloud–Edge Application Deployment	· Successfully deploys and demonstrates an application across cloud and edge environments. · Effectively analyzes workload placement and communication between device, edge, and cloud. · Clearly justifies deployment decisions.	· Deploys the application with some limitations. · Demonstrates basic understanding of cloud-edge workload placement.	· Unable to properly deploy or analyze the cloud-edge application. · Limited understanding of workload placement.	3.2.3 / 5.2.1	5
AI Workload Analysis and Dynamic Computation Offloading	· Effectively evaluates AI workload performance under varying load conditions. · Successfully designs and analyzes dynamic computation offloading across device, edge, and cloud. · Clearly compares performance and resource utilization.	· Performs basic AI workload evaluation and computation offloading. · Analysis contains minor gaps or limitations.	· Unable to properly evaluate workload performance or implement computation offloading. · Analysis is incomplete or incorrect.	3.2.2 / 5.2.1	5

Date **Head, CSE(AI)**

Chapter wise Plan

Course Code and Title: Cloud Computing	**26ECAC301**
Chapter Number and Title: 1. Introduction	**Planned Hours: 8hrs**

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the motivation, characteristics, scalability, and elasticity of cloud computing.	CO1	L3	1.4
2. Differentiate centralized and distributed computing models and various cloud service and deployment models.	CO1	L3	1.4
3. Explain the role of edge computing, data centers, and virtualization in reducing latency and bandwidth usage.	CO1	L2	1.4
4. Analyze cost and performance trade-offs between Cloud Computing environments.	CO1	L3	2.1

Lesson Schedule Class No. Portion covered per hour
1 Introduction to Cloud Computing: Motivation, need for cloud computing, characteristics, and benefits
2 Elastic Computing and Scalability: Elasticity, scalability, resource provisioning, and practical examples
3 Computing Models: Centralized and distributed computing models, comparison and applications
4 Cloud Service Models: IaaS, PaaS, SaaS, their characteristics, examples, and applications
5 Cloud Deployment Models: Public, private, hybrid, and community cloud models
6 Introduction to Edge Computing: Motivation, need for edge computing, and role in reducing latency and bandwidth usage
7 Data Center and Virtualization: Data center architecture, basic virtualization concepts, and their role in cloud computing
8 Cloud–Edge Trade-offs: Cost and performance comparison, latency, bandwidth, resource utilization, and suitable use cases

Review Questions

Sl. No.	Questions	TLOs	BL	PI Code
1.	Discuss the benefits of migrating e-commerce business to a Cloud Computing infrastructure. What are key considerations considered during the transition process?	TLO1	L3	1.4.3
2.	How does elastic computing differ from traditional fixed-capacity infrastructure models, and what are the primary advantages it offers.	TLO1	L2	1.4.3

3.	Assume you are a strategic consultant advising a group of investors interested in entering the Cloud Computing market as service providers. They are seeking in-depth insights into various business models that have proven successful in this sector.	TLO2	L3	1.4.3
4.	Apply the concepts of centralized and distributed computing to explain how cloud computing supports scalable application deployment.	TLO2	L3	1.4.3
5.	Considering the importance of reliability and uptime for a startup's cloud-based services, what strategies would you employ to ensure high availability and fault tolerance within the network infrastructure? How can redundancy and failover mechanisms be effectively implemented within the leaf-spine topology to minimize service disruptions and maintain uninterrupted access to critical resources for both internal users and external clients?	TLO3	L3	1.4.3
6.	A smart city generates large amounts of data from traffic cameras and IoT sensors. Sending all data to a distant cloud causes high latency and increased bandwidth usage. Apply edge computing concepts to propose suitable cloud–edge architecture and explain how it can improve system performance.	TLO4	L3	1.4.3
7.	A healthcare organization needs to deploy applications for patient data management, medical image processing, and remote monitoring. Analyze the requirements and recommend suitable cloud service and deployment models for each application. Justify your choices based on security, scalability, cost, and accessibility.	TLO3	L3	1.4.3
8.	A video-streaming company experiences fluctuating user demand, with a significant increase in traffic during weekends and special events. Apply the concepts of elasticity and scalability to propose a suitable resource-provisioning strategy. Explain how the strategy can maintain performance while minimizing unnecessary resource costs.	TLO4	L3	1.4.3

Chapter wise Plan

Course Code and Title: Cloud Computing	26ECAC301
Chapter Number and Title: 2. Virtualization and Containerization.	Planned Hours: 7hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain virtualization approaches, types of hypervisors, virtualization models, and containerization concepts.	CO2	L3	2.1
2. Demonstrate container execution, operating system isolation, networking, and persistent storage mechanisms.	CO2	L2	1.4
3. Compare virtual machines and containers based on performance, resource utilization, isolation, scalability, and deployment requirements.	CO2	L3	3.2

Lesson Schedule Class No. - Portion covered per hour
1 Virtual Machines: Concept of virtualization, need for virtualization, and virtual machine architecture
2 Virtualization Approaches: Types of hypervisors, virtualization models, and their applications
3 Containerization: Container concepts, benefits, and comparison with virtual machines
4 OS Isolation & Runtime: Operating system isolation mechanisms, container execution environment, and runtime abstraction
5 Container Networking: Container networking concepts, communication between containers, and network configuration
6 Container Storage: Persistent storage, volumes, and data management for containerized applications
7 VMs vs Containers: Comparison based on performance, resource utilization, isolation, scalability, deployment, and associated trade-offs

Review Questions

Sl. No.	Questions	TLOs	BL	PI Code
1.	Discuss the primary types of virtualization approaches.	TLO1	L2	2.1.2
2.	Analyze the security measures which can be implemented to ensure trust within virtual machines operating on the same physical host?	TLO1	L3	2.1.2
3.	Discuss the advantages and disadvantages of virtual machines in this enterprise environment. Additionally, propose how isolation facilitates within an operating system can complement the security measures.	TLO2	L3	1.4.3
4.	In a Cloud Computing environment, a software development company seeks to deploy multiple applications with different dependencies on the same Linux- based server. They want to ensure strong isolation between these applications to prevent conflicts and enhance security. Explain how Linux namespaces can be utilized for isolation in this scenario. Additionally, propose a container-based approach to achieve isolated application deployment.	TLO2	L3	2.1.2
5.	A data center manages a monolithic application deployed across multiple servers, resulting in high resource consumption and operational challenges. The IT team considers containerization using Docker to streamline deployment and management. Explain how Docker containers can address the challenges faced by the data center with the monolithic application. Additionally, outline the key software components of Docker and mention some items typically found in a Docker file.	TLO3	L3	3.2.2
6.	A cloud application consists of multiple containerized services that need to communicate with each other while maintaining network isolation. Apply suitable container networking concepts to design the communication among these services. Explain the role of network configuration in enabling secure and efficient communication.	TLO2	L3	1.4.3
7.	A company is migrating an application from virtual machines to containers to improve deployment speed and resource utilization. Analyze the requirements of the application and recommend whether VMs or containers should be used. Justify your choice based on performance, resource utilization, isolation, scalability, and deployment considerations.	TLO3	L3	1.4.3

Chapter wise Plan

Course Code and Title: Cloud Computing	26ECAC301
Chapter Number and Title: 3. Automation and Orchestration.	Planned Hours: 8hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain orchestration principles and Kubernetes architecture, including nodes, pods, and the control plane.	CO3	L3	3.2
2. Demonstrate container lifecycle management, scheduling, resource allocation, and workload placement across nodes.	CO3	L2	3.2
3. Apply Kubernetes-based edge architectures, infrastructure-as-code concepts, and automation tools for efficient application management.	CO3	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1 Orchestration Principles: Need for automation and orchestration, key principles, and benefits
2 Kubernetes Architecture: Kubernetes components, architecture, and working principles
3 Kubernetes Cluster Model: Control plane, nodes, pods, and their roles
4 Container Lifecycle Management: Container creation, deployment, scaling, updating, and termination
5 Scheduling and Resource Allocation: Pod scheduling, resource requirements, allocation, and utilization
6 Workload Placement: Placement of workloads across nodes and factors affecting scheduling decisions
7 Kubernetes at the Edge: Kubernetes-based edge architectures and challenges in edge environments
8 Infrastructure Automation: Infrastructure as Code concepts and automation tools for managing cloud and edge infrastructure

Review Questions

Sl. No.	Questions	TLOs	BL	PI Code
1.	Explain the concept of automation in data centers and discuss the different levels of automation that can be implemented in this scenario.	TLO1	L2	3.2.2
2.	A large-scale cloud provider aims to streamline its infrastructure deployment and management processes to meet growing demands efficiently. They seek to implement advanced automation strategies to achieve this goal. Describe the concepts of zero-touch provisioning and Infrastructure as Code (IaC) in the context of automation. Additionally, discuss the role of orchestration tools in enabling automation with a broader scope in this scenario.	TLO1	L3	3.2.2
3.	A technology company plans to deploy its microservices-based applications using Kubernetes to improve scalability and reliability. They are exploring the Kubernetes cluster model to understand how it can facilitate efficient management of their containerized workloads. Illustrate the Kubernetes cluster model and its components using an example. How does this model contribute to the effective orchestration of containerized applications?	TLO2	L3	5.2.1
4.	A cloud-native development team is transitioning their microservices-based applications to Kubernetes for improved scalability and manageability. In the context of Kubernetes, elaborate on the creation process, templates, and binding time of pods.	TLO3	L3	3.2.2
5.	A cloud application experiences varying workloads during the day. Apply Kubernetes scheduling and resource allocation concepts to deploy and distribute pods across the available nodes based on their CPU and memory requirements.	TLO2	L3	1.4.3
6.	A company has a Kubernetes cluster with nodes having different CPU, memory, and storage capacities. Apply suitable workload placement techniques to deploy different application workloads on appropriate nodes and explain the factors considered during placement.	TLO3	L2	5.2.1
7.	A company wants to deploy applications using Kubernetes at remote edge locations with limited computing resources and intermittent connectivity. Apply Kubernetes-at-the-edge concepts to design a suitable deployment approach and explain how it addresses the resource and connectivity constraints.	TLO2	L3	1.4.3
8.	A development team frequently configures cloud and edge infrastructure manually, leading to inconsistent deployments. Apply Infrastructure as Code (IaC) principles to automate the provisioning and configuration of the infrastructure. Illustrate the steps involved in the proposed approach.	TLO3	L3	3.2.2

Chapter wise Plan

Course Code and Title: Cloud Computing	26ECAC301
Chapter Number and Title: 4. Microservices and Distributed Applications	Planned Hours: 7hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the architecture, characteristics, advantages, limitations, and service granularity of microservices-based applications.	CO4	L2	1.4
2. Apply service discovery, load balancing, stateless servers, and containerization for microservice deployment.	C04	L3	3.2
3. Design and deploy microservices across cloud and edge environments while addressing distributed system challenges.	CO4	L3	3.2
4. Explain and apply serverless computing concepts and serverless infrastructure architecture for distributed applications , observability, and reliability among microservices.	CO4	L3	5.2

Lesson Schedule Class No. - Portion covered per hour
1. Microservices Architecture, Advantages, Limitations, and Service Granularity
2. Service Discovery and Load Balancing
3. Stateless Servers and Containers
4. Development of Microservices
5. Deployment of Microservices in Cloud and Edge Environments
6. Challenges in Distributed System Design
7. Serverless Computing and Serverless Infrastructure Architecture

Review Questions

Sl. No	Questions	TLOs	BL	PI Code
1.	A software development company is considering adopting microservices architecture for their next project. Discuss the advantages and disadvantages of adopting microservices architecture in the context of software development.	TLO1	L3	1.4.3
2.	Assume that a software development team is designing a new system architecture for their e-commerce platform. They are deliberating on the granularity of microservices and the communication protocols suitable for their architecture. Discuss the concept of granularity in microservices architecture and elucidate the communication protocols commonly used for microservices-based systems.	TLO2	L3	3.2.2
3.	Explain how microservices architecture improves scalability, flexibility, and independent deployment compared with a monolithic architecture.	TLO2	L2	3.2.2
4.	Analyse the serverless computing approach to design a simple application that processes user-uploaded images. Identify suitable serverless functions and explain how they would be triggered and executed.	TLO3	L3	3.2.2
5.	Evaluate how service discovery and load-balancing mechanisms improve the scalability, availability, and reliability of distributed applications.	TLO3	L2	5.2
6.	An e-commerce application consists of separate microservices for payment, inventory, order processing, and user management. During a festival sale, some services receive significantly higher requests than others. Analyze how service discovery and load balancing can be applied to manage the increased workload and maintain service availability.	TLO4	L3	5.2
7.	Apply the serverless infrastructure architecture to develop a cloud-based application for processing IoT sensor data. Identify the major components and explain the flow of data through the serverless system.	TLO5	L3	2.1

Chapter Number and Title: 5. Edge-Cloud Continuum	Planned Hours: 5hrs

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain the fundamentals of edge computing and the edge–cloud continuum, including the device–edge–cloud hierarchy.	CO4	L2	1.4
2. Analyze latency-sensitive and real-time applications to determine suitable application placement and computation offloading strategies across devices, edge, and cloud.	CO4	L3	3.2
3. Evaluate the impact of data locality, communication overhead, and latency on system performance and apply latency-aware design principles.	CO4	L3	3.2
4. Apply edge–cloud continuum concepts to real-world AI, IoT, and real-time system use cases, considering latency, resource availability, and computational requirements.	CO4	L3	5.2

Lesson Schedule Class No. - Portion covered per hour
1. Edge Computing Fundamentals and Characteristics
2. Edge–Cloud Continuum and Device–Edge–Cloud Hierarchy
3. Latency-Sensitive and Real-Time Applications
4. Application Placement Strategies Across Edge and Cloud
5. Computation Offloading Basics and Techniques
6. Data Locality, Communication Overhead
7. Latency-Aware Design
8. Edge–Cloud Use Cases in AI, IoT and Real-Time Systems

Review Questions

Sl.No.	Questions	TLOs	BL	PI Code
1.	Analyze different application placement strategies for deploying applications across devices, edge nodes, and cloud servers based on latency and resource requirements.	TLO1	L3	1.4.3
2.	Explain the concept of the Edge–Cloud Continuum and describe the device–edge–cloud hierarchy with a suitable example.	TLO2	L2	3.2.3

3	Describe computation offloading and explain the importance of data locality and communication overhead in edge–cloud systems.	TLO1	L2	3.2.3
4.	Analyze how latency-aware system design can improve the performance of latency-sensitive and real-time applications in an Edge–Cloud Continuum.	TLO3	L3	3.2.3
5.	A smart traffic management system collects data from thousands of roadside IoT sensors. The system must process emergency traffic events with very low latency, while historical traffic data requires large-scale processing. Analyze how tasks can be distributed among IoT devices, edge nodes, and cloud servers to meet these requirements.	TLO4	L3	5.2.1
6.	An AI-based healthcare monitoring application continuously collects data from wearable devices. Critical health events must be detected immediately, while long-term patient data is analyzed periodically. Apply suitable computation offloading and application placement strategies across the device–edge–cloud hierarchy to reduce latency and communication overhead.	TLO4	L3	5.2.1
7.	A smart manufacturing system uses cameras and sensors to monitor machines in real time. Apply latency-aware design and edge computing concepts to determine which processing tasks should be performed at the devices, edge nodes, and cloud servers. Justify the placement based on latency, bandwidth, and computational requirements.	TLO3	L3	3.2.3
8.	An autonomous warehouse uses robots that continuously generate sensor and video data. Apply computation offloading and data locality concepts to design a suitable edge–cloud processing approach. Explain how your approach can reduce communication overhead and support real-time decision-making.	TLO4	L2	5.2.1

Course Code and Title: Cloud Computing	**26ECAC301**
Chapter Number and Title: 6. Resource Management	**Planned Hours: 5hrs**

Learning Outcomes: -

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	CA Code
1. Explain resource allocation strategies for efficient utilization of computing, storage, and network resources.	CO5	L2	3.2
2. Analyze task scheduling and load-balancing techniques for distributed systems.	CO5	L3	3.2
3. Evaluate distributed system performance using latency, throughput, response time, resource utilization, and energy consumption metrics.	CO5	L3	5.2
4. Apply optimization strategies to improve performance, scalability, and resource efficiency in large-scale distributed systems.	CO5	L3	5.2

Lesson Schedule Class No. - Portion covered per hour
1. Resource Management Fundamentals and Resource Allocation Strategies
2. Resource Allocation Techniques in Distributed Systems
3. Task Scheduling Strategies and Scheduling Algorithms
4. Load Balancing Techniques and Workload Distribution
5. Performance Evaluation Metrics and System Monitoring
6. System Optimization Strategies for Distributed Systems
7. Large-Scale Distributed System Optimization and Case Studies

Review Questions

Sl. No.	Questions	TLOs	BL	PI Code
1.	Explain different resource allocation strategies used in distributed systems and discuss their importance.	TLO1	L2	3.2.3
2.	A cloud–edge application experiences uneven workload distribution, where some edge nodes are overloaded while others remain underutilized. Apply suitable task scheduling and load-balancing strategies to distribute workloads efficiently and improve system performance.	TLO2	L3	3.2.3
3.	Describe task scheduling and load balancing techniques used for efficient resource utilization in distributed sy stems.	TLO3	L3	3.2.3

4.	A large-scale IoT application processes thousands of tasks across multiple edge and cloud servers. The system is experiencing high latency and poor resource utilization. Analyze the situation and propose suitable resource allocation and optimization strategies to improve performance and scalability.	TLO4	L3	5.2.1
5.	Analyze how performance evaluation metrics such as latency, throughput, response time, resource utilization, and energy consumption can be used to evaluate a distributed system.	TLO4	L3	5.2.1
6.	Evaluate different system optimization strategies and determine how they can improve scalability, resource utilization, and overall system performance.	TLO4	L3	5.2.1

Model Question Paper for Minor Examination (ISA-I)						
Course Code: 26ECAC301		**Course Title: Cloud Computing**				
Duration: 75 Mins		**Max. Marks: 40**				
Note: Answer any 2 full Questions. Each full question carries 15 marks						
Q.No	**Questions**	**Marks**	**CO**	**BL**	**PO**	**PI Code**
1.a	Explain the concept of **v**irtualization in cloud computing. Describe Type-1 and Type-2 hypervisors and compare their characteristics with suitable examples.	6	CO1	L2	1	1.4.3
1.b	A college plans to deploy an online learning platform that requires infrastructure management, application development support, and access to ready-to-use software services. Identify and differentiate IaaS, PaaS, and SaaS, and explain which service model is suitable for each requirement.	6	CO2	L3	2	2.1.2
1.c	A smart traffic management system collects data continuously from cameras and sensors deployed at different locations. Sending all the data to a centralized cloud increases network bandwidth usage and causes delays in detecting traffic congestion. The system requires quick responses for real-time traffic monitoring.Analyze how an edge–cloud architecture can address this problem. Explain the placement of processing at the edge and cloud, and evaluate the solution using latency, bandwidth usage, resource utilization, and response time.	8	CO1	L3	1	1.4.3
2.a	Explain the concept of virtualization in cloud computing. Describe Type-1 and Type-2 hypervisors and compare their characteristics with suitable examples.	6	CO2	L2	2	2.1.2
2.b	A cloud provider needs to run multiple applications on a single physical server while maintaining separate computing environments for each application. Identify the virtualization approach required and explain how virtual machines, hypervisors, CPU, memory, and storage resources are managed in this environment.	6	CO3	L3	3	3.2.2
2.c	An organization has limited physical servers but needs to deploy several applications with different CPU, memory, and storage requirements. Each application must operate independently without affecting other applications. The organization also wants better resource utilization and reduced infrastructure cost.	8	CO3	L2	2	2.1.2

	Analyze how virtualization can be used to address these requirements. Discuss the role of hypervisors and evaluate the solution in terms of resource utilization, isolation, performance, scalability, and cost.					
3.a	Explain the concept of containerization and describe the operating-system-level isolation mechanisms used by containers. Explain the role of the container execution environment and runtime abstraction.	6	CO3	L2	3	3.2.2
3.b	How can a medium-scale e-commerce company strategically leverage various cloud deployment models to enhance operational flexibility and mitigate risks associated with vendor lock-in? How does adopting a multi-cloud approach contribute to optimizing performance and resilience?	6	CO1	L3	1	1.4.3
3.c	A software company wants to deploy multiple independent application services. The services should start quickly, use resources efficiently, communicate through a network, and retain data even when containers are stopped or replaced. The company is considering both virtual machines and containers for deployment. Analyze the suitability of containers for this application. Compare containers and virtual machines with respect to isolation, startup time, resource utilization, networking, persistent storage, and performance, and justify the suitable approach.	8	CO2	L3	3	3.2.2

Model Question Paper for Minor Examination (ISA-II)	
Course Code: 26ECAC301	**Course Title: Cloud Computing**
Duration: 75 Mins	**Max. Marks: 40**
Note: Note: Answer any 2 full Questions. Each full question carries 15 marks	

Q. No	Questions	Marks	CO	BL	PO	PI Code
1.a	Explain the principles of container orchestration. Describe the major components of Kubernetes architecture, including the control plane, nodes, and pods..	6	CO4	L2	1	1.4.3
1.b	A Kubernetes cluster consists of multiple worker nodes with different CPU and memory capacities. Explain how Kubernetes scheduling and resource allocation can be used to place workloads across these nodes. Discuss the factors considered during workload placement.	6	CO4	L3	3	3.2.2
1.c	A college deploys a web application on a Kubernetes cluster consisting of multiple worker nodes. During peak hours, some nodes become heavily loaded while other nodes have sufficient resources. The application should remain available even when a node becomes unavailable. Analyze how Kubernetes scheduling, resource allocation, and workload placement can be used to manage this application. Explain how the Kubernetes cluster can improve resource utilization, availability, and scalability.	8	CO4	L3	2	2.1.2
2.a	Explain the container lifecycle management process in Kubernetes. Describe how Kubernetes manages the creation, execution, scaling, and termination of containers.	6	CO4	L2	1	1.4.3
2.b	Explain the concept of Infrastructure as Code (IaC). Discuss how automation tools can be used to provision and manage cloud infrastructure with reduced manual intervention.	6	CO5	L2	3	3.2.3
2.c	A smart manufacturing system has several edge devices that generate data and require applications to be deployed close to the machines. The system has limited resources at edge nodes, while additional computing resources are available in the cloud. Applications may need to be moved or deployed at different locations based on workload. Analyze how a Kubernetes-based edge architecture can be used for this system. Discuss workload placement, resource allocation, scalability, and automation, and	8	CO4	L3	5	5.2.1

	explain how IaC can reduce manual configuration.					
3. a	Explain microservices architecture and discuss its major advantages and limitations. Explain how service granularity affects the design of a microservices-based application.	6	CO5	L2	3	3.2.3
3.b	A web application consists of multiple independent services that need to communicate with each other. Explain the role of service discovery and load balancing in such a distributed application. Also explain why stateless servers and containers are commonly used.	6	CO4	L2	3	3.2.2
3.c	An e-commerce application consists of separate services for user management, product catalog, payment, and order processing. During a festival sale, the number of requests increases significantly, and some services receive much higher traffic than others. The application must continue operating even if one service temporarily fails. Analyze how microservices architecture can be designed for this application. Discuss service granularity, service discovery, load balancing, independent scaling, and fault handling.	8	CO4	L3	1	1.4.3

Model Question Paper for End Semester Assessment						
Course Code: 26ECAC301		**Course Title: Cloud Computing**				
Duration: 3 hrs						
Max. Marks: 100						
Note: i) Answer any TWO full questions from UNIT-I, any TWO full questions from UNIT-II , answer any 1 from Unit-III. **ii) Assume suitable data if necessary with justification**						
	UNIT-I					
Q.No	**Questions**	**Marks**	**CO**	**BL**	**PO**	**PI Code**
1a	Amazon Web Service (AWS) is a leading cloud service provider that wants to set up a new data center in Hubballi. AWS has assigned you to set up that data center using leaf-spine network topology. How do you implement the same to enhance network performance scalability and address issues like congestion and bandwidth limitations?	8	CO1	L3	1	1.4.3
1b	A hospital's IT department is managing a private cloud infrastructure that hosts virtual machines (VMs) for critical applications, including an AI-powered medical image analysis system. Due to an upcoming maintenance operation on one of the physical servers, the team needs to migrate a running VM to another host without disrupting the AI system's real-time processing of medical images. Describe the process of VM live migration for the scenario mentioned.	6	CO2	L3	2	2.1.2
1c	Analyze the trade-offs between cloud computing and edge computing in terms of latency, bandwidth usage, cost, and performance.	6	CO1	L2	3	3.2.2
2a	Describe Docker container Technology; also explain its main software components and the key elements of a Dockerfile. How do these components support containerized applications' creation, deployment and management in modern software development?	8	CO2	L2	2	2.1.2
2b	Digital Realty: A multinational corporation operates multiple data centers across regions to support its cloud-based services. Data center operations, including workload distribution, power management, cooling, and security, are handled with manual and semi- automated processes. How will this company adopt different levels	6	CO3	L3	3	3.2.2

	of automation techniques to improve efficiency, reduce operational costs, and enhance reliability?					
2c	A cloud provider needs to run applications belonging to multiple customers while ensuring isolation, efficient resource utilization, and easy application deployment. Analyze whether virtual machines or containers would be more suitable. Justify your choice based on isolation, overhead, resource utilization, startup time, and portability.	6	CO3	L3	2	2.1.2
3a	The adoption of Kubernetes as the container orchestration system outlines the key challenges and benefits faced by the organization in managing containerized applications. Discuss how Kubernetes addresses these challenges and facilitates efficient resource utilization.	8	CO3	L2	3	3.2.2
3b	How can a medium-scale e-commerce company strategically leverage various cloud deployment models to enhance operational flexibility and mitigate risks associated with vendor lock-in? How does adopting a multi-cloud approach contribute to optimizing performance and resilience?	6	CO1	L3	1	1.4.3
3c	An edge computing system has three nodes with different CPU and memory capacities. A new application requires high CPU resources, while another app4lication requires high memory resources. Apply Kubernetes scheduling and workload-placement concepts to determine how these workloads should be distributed across the nodes. Explain how resource requirements can influence the scheduling decision.	6	CO2	L3	2	2.1.2
	UNIT-II					
4.a	An e-commerce application is currently developed as a monolithic application. As the number of users increases, updating one feature requires redeploying the entire application. Analyze how a microservices architecture could solve this problem. Explain how service granularity, inter-service communication, service discovery, and independent deployment can be applied.	8	CO4	L3	1	1.4.3
4.b	A smart factory generates video data from cameras that must be processed within a few milliseconds, while historical production data can be analyzed later. Analyze how the device–edge–cloud hierarchy can be used to place these workloads. Justify which tasks should execute at the device, edge, and cloud based on latency, data locality, and communication overhead.	6	CO4	L3	3	3.2.2

4.c	Apply task scheduling and resource allocation strategies to efficiently distribute workloads across multiple edge nodes.	6	CO3	L2	2	2.1.2
5.a	A large-scale IoT application generates tasks with different execution times, resource requirements, and deadlines. Some tasks are latency-sensitive while others can tolerate delays. Apply task-scheduling and resource-allocation strategies to determine how these tasks should be assigned across distributed edge nodes. Explain how your approach can optimize latency, resource utilization, and overall system performance.	8	CO4	L3	1	1.4.3
5.b	Apply workload-placement concepts to distribute applications across heterogeneous nodes.	6	CO5	L3	3	3.2.3
5.c	Explain the role of Infrastructure as Code (IaC) in automation.	6	CO4	L2	1	1.4.3
6.a	An edge application experiences rapidly changing workloads. Analyze how Kubernetes orchestration can be used to manage container lifecycle, scheduling, and resource allocation.	8	CO5	L3	3	3.2.3
6.b	A company has three Kubernetes nodes with different CPU and memory capacities. Apply Kubernetes scheduling concepts to determine how workloads should be placed on these nodes.	6	CO4	L3	3	3.2.2
6.c	Describe different inter-service communication mechanisms.	6	CO4	L2	2	2.1.2
Unit-III						
7.a	An IoT application generates a large amount of sensor data, but only a small portion requires immediate processing. Apply computation offloading to decide which data should be processed at the edge and which should be sent to the cloud.	8	CO3	L3	1	1.4.3
7.b	Explain the concept of application placement in an edge–cloud continuum.	6	CO4	L2	2	2.1.2
7.c	Consider a set of computational tasks with different CPU and memory requirements and multiple edge nodes with varying resource capacities. Apply a suitable task-scheduling strategy to distribute the tasks among the edge nodes. Show the task allocation and explain how the selected strategy improves resource utilization and reduces task execution time.	6	CO4	L3	1	1.4.3
8.a	An edge node has limited CPU and memory, while incoming applications have different resource requirements. Apply resource allocation strategies to	8	CO5	L3	3	3.2.3

	determine how the available resources should be distributed.					
8.b	Explain the need for resource management in distributed edge-cloud systems.	6	CO4	L2	3	3.2.2
8.c	An autonomous vehicle requires responses within milliseconds. Analyze how an edge–cloud continuum can be designed to satisfy the application's latency requirements.	6	CO4	L3	2	2.1.2